

Translated

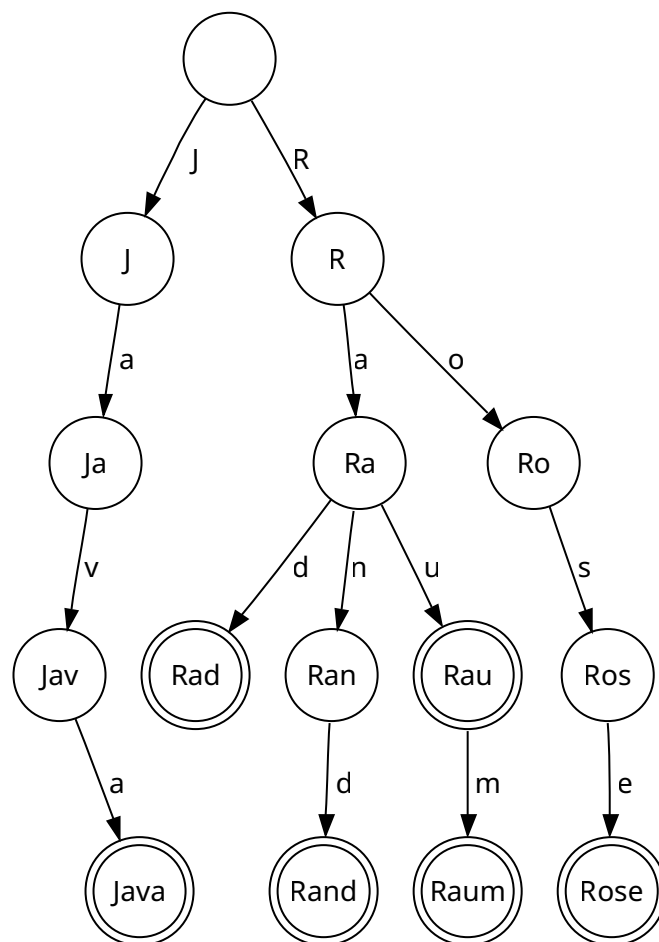
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Aho-Corasick algorithm

The Aho-Corasick algorithm allows us to quickly search for multiple patterns in a text. The set of pattern strings is also called a *dictionary*. We will denote the total length of its constituent strings by m and the size of the alphabet by k . The algorithm constructs a finite state automaton based on a trie in $O(mk)$ time and then uses it to process the text.

The algorithm was proposed by Alfred Aho and Margaret Corasick in 1975.

Construction of the trie



A trie based on words "Java", "Rad", "Rand", "Rau", "Raum" and "Rose".